DEFENCE AI-ANC SYSTEM

Hardware & Embedded Systems Summary

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1. What Is This Project?

This project is a custom, military-grade circuit board with a built-in Artificial Intelligence chip. Its purpose is to perform **Active Noise Cancellation (ANC)** in real-time for soldiers and pilots in extreme environments.

Unlike standard noise-canceling headphones, this board uses a Deep Learning AI model specifically trained to erase loud, chaotic battlefield noises—such as **gunshots (165 dB), artillery blasts, and helicopter rotor wash**—so that voice communication remains crystal clear.

The entire process, from capturing the noise to outputting clean audio, happens in just **18 milliseconds**, so there is no noticeable delay.

2. How It Works (The Signal Flow)

The system processes audio in a straight line from input to output:

- **1. Audio Input:** Two heavy-duty military microphones (capable of surviving 141 dB of sound) capture the voice and the background noise.
- **2. Pre-Amplifier:** An ultra-low-noise amplifier boosts the tiny microphone signals so they can be read accurately.
- **3. Digitization (ADC):** A 24-bit Analog-to-Digital Converter turns the audio into digital data that the computer chips can understand.
- **4. Central Processor (MCU):** An STM32 processor collects the data and instantly routes it to the AI chip.
- **5. AI Noise Removal (NPU):** The Kendryte K210 AI Chip runs our custom neural network. It isolates human speech and deletes the battlefield noise in just 12 milliseconds.
- **6. Audio Output (DAC & Amp):** The cleaned digital signal is converted back to analog audio, and a powerful 315W amplifier drives it into the user's headset.

3. Detailed Bill of Materials (BOM) & Costing

To build the first prototype, the total cost is approximately **$224.20 per unit**. When manufactured at scale (1,000+ units), the cost drops to about **$85.00 per unit**.

Below is the complete list of major active components used on the board, their exact part numbers, functions, and estimated prototype costs:

Ref	Part Number	Description	Est. Cost
PCB	Custom 4-Layer	FR4 Tg170, ENIG Finish (160x100mm)	$45.00
U1	STM32H755ZIT6	Dual-Core Central Microcontroller (MCU)	$18.50
U2	Kendryte K210	Artificial Intelligence Chip (NPU)	$8.00
U14	IS42S32200L	64Mb SDRAM (Memory for AI Models)	$4.50
U13	W25Q128	128Mb SPI Flash (Storage)	$1.80
MK1,2	ICS-40300 (x2)	Military MEMS Microphones (141dB SPL)	$2.40
U4	INA217	Low-Noise Microphone Pre-Amplifier	$6.80
U18	MAX7401	8th-Order Low-Pass Filter	$4.10
U5	ADS1274	24-bit 144kSPS Analog-to-Digital Converter	$28.00
U3	CS4270	Digital Audio Codec	$8.50
U10	PCM1794A	24-bit Digital-to-Analog Converter (DAC)	$9.80
U9	TPA3255	315W Class-D Audio Power Amplifier	$12.50
U6	TPS65217	Power Management IC (PMIC)	$4.20
U7, U8	TLV1117 Series	3.3V and 1.8V Linear Voltage Regulators	$1.70
U11	SP3485	RS-485 Communication Transceiver	$2.50
U12	TCAN1044	CAN FD Communication Transceiver	$2.50
U15	ICM-42688	6-Axis IMU (Vibration Tracking)	$2.90
U16	BMP388	Barometric Pressure Sensor (Wind Tracking)	$2.20
Misc	Passives/Conn	Resistors, Capacitors, Connectors	$18.30

4. Military Durability & Standards

Because this hardware is used in defense applications, it is built to survive extreme conditions:

- **MIL-STD-810H:** Rated to survive temperatures from -40°C to +85°C, as well as thermal shock, extreme humidity, and heavy vibration.
- **IP67 Waterproofing:** The entire circuit board is housed inside a sealed Hammond aluminium enclosure with waterproof military connectors.
- **Conformal Coating:** After the chips are soldered on, the board is sprayed with a special chemical layer (Humiseal 1A33) that prevents dust or condensation from causing electrical shorts.
- **ITAR Controlled:** The design contains restricted defense technology, so the board itself has 'ITAR CONTROLLED' printed on it in silkscreen.